

# ES 9.x Migration — Requirements Dashboard

migration\_v0HLD

migration\_v1DRAFT

P1 Write consolidationANALYSIS

HLD committed path (P0, P2–P4) · P1 consumer consolidation is draft · 3M MAU cohort · dual-write · experiment reads on C3/C4

Search tasks, files, repos...

PHASE

All · None

☒ P0

☒ P1

☒ P2

☒ P3

☒ P4

☒ P5

☒ P6

CATEGORY

All · None

☒ library

☒ service

☒ infra

ROLE

All · None

☒ query

☒ adapter

☒ models

☒ write

☒ read

☒ infra

STATUS

☒ All

☐ Pending

☐ Done

- ✓ elasticquery-dsl-py
- ✓ elastictoolkit-py
- ✓ wi-elasticsearchmodels-py
- ✓ wicandidateprofile-ms
- ✓ wi-candidateevents-ms
- ✓ wiemployerviewcandidate-ms
- ✓ wiemployerjobpost-ms
- ✓ workindiajobs-ms
- ✓ wiemployerservicingms
- ✓ wirelevancyadapter (external)
- ✓ NEW: single-write-consumer
- ✓ NEW: embedding-pipeline
- ✓ Infrastructure / Ops

### migration\_v0 — HLD Committed Path

Semantic matching experiment on 3M MAU. Production reads stay on C1/C2 until conditional P6 cutover.

- P1 (DRAFT): Consolidate 5 writers → single Kafka consumer — not confirmed yet; alternative write paths under review
- P2: Write freeze <1hr → \_reindex MAU cohort → verify → resume dual-write
- P3: Embedding pipeline (~9.4M vectors) — critical path
- P4: Experiment kNN reads on C3/C4 only — not production Consul
- 404 on C3 is write-path backfill, not a production read fallback

**13**

Repositories

**62**

Total tasks

**0**

Completed

**62**

Pending

**0%**

Progress

[HLD — Architecture](#)[HLD — Roadmap](#)[HLD — Migration Window](#)[HLD — Dual-Write Flow](#)[C1 — Candidate ES](#)[C2 — Jobs ES](#)

ES 8.10.0

Source of record · 47.1M docs · all prod candidate reads

ES 8.6.0

Source of record · ~150K jobs · 300–400 q/s

### C3 — Candidate ES (experiment)

ES 9.4.2

workindia-candidate-feature-embedded · 3M MAU first

### C4 — Jobs ES (experiment)

ES 9.4.2

Job context vectors for Candidate→Job kNN

All phases

P0 Pre-flight

P1 Consumer consolidation (DRAFT)

P2 Migration window

P3 Embeddings

P4 Experiment

P5 Remaining ~44M

P6 Production cutover

## Write Fragmentation (P1 draft — not confirmed)

**ES indices** — how many distinct Elasticsearch indexes this service writes to (e.g. candidate feature vs job-post status).  
**Write paths** — Kafka consumers plus HTTP, cron, or other triggers that call ES; the breakdown in parentheses shows the split. Expand a card for field groups, files, and each index.

+ **wicandidateprofile-ms** **WRITER** 1 ES index · 15 write paths — 13 Kafka consumers + 2 HTTP/cron/other

+ **wi-candidateevents-ms** **WRITER** 1 ES index · 6 write paths — 3 Kafka consumers + 3 HTTP/cron/other

+ **wiemployerviewcandidate-ms** **WRITER** 4 ES indices · 8 write paths — 5 Kafka consumers + 3 HTTP/cron/other

+ **wiemployerjobpost-ms** **WRITER** 1 ES index · 8 write paths — 7 Kafka consumers + 1 HTTP/cron/other

+ **workindiaapi (shared)** **WRITER** 1 ES index · 4 write paths — 4 HTTP/cron/other

## Read Fragmentation · production on C1/C2 through P4

**ES indices** — how many distinct Elasticsearch indexes this service reads from. **Read paths** — Kafka consumers plus HTTP, cron, or library calls that issue search/get/mget; the breakdown in parentheses shows the split. Expand a card for fields queried, files, and each index.

+ **workindiajobs-ms** **READER** 2 ES indices · 15 read paths — 9 Kafka consumers + 6 HTTP/cron/other

+ **wi-candidateevents-ms** **READER**

1 ES index · 1 read path — 1 HTTP/cron/other

+ **wiemployerviewcandidate-ms**  
**READER**

3 ES indices · 9 read paths — 3 Kafka consumers + 6 HTTP/cron/other

+ **wicandidateprofile-ms** **READER**

2 ES indices · 7 read paths — 4 Kafka consumers + 3 HTTP/cron/other

+ **wiemployerjobpost-ms** **READER**

3 ES indices · 5 read paths — 2 Kafka consumers + 3 HTTP/cron/other

+ **wiemployerservicingms** **READER**

1 ES index · 4 read paths — 2 Kafka consumers + 2 HTTP/cron/other

+ **workindiaapi (legacy)** **READER**

1 ES index · 3 read paths — 3 HTTP/cron/other

## Libraries

+ **elasticquery-dsl-py**

Low-level ES query JSON builder — no client dependency · pinned: 1.1.0 · click to expand

library

query

0/3

+ **elasticsearchtoolkit-py**

ES client adapters + directive query engine · pinned: 0.8.0 · click to expand

library

adapter

0/5

+ **wi-elasticsearchmodels-py**

Shared index models, cluster adapters, field enums · pinned: 1.1.1 · click to expand

library

models

0/8

+ **wirelevancyadapter (external)**

CandidateJobOrderGenerationES8 — query building + execute() for jobs-ms · pinned: external PyPI · click to expand

library

query

0/2

## Services

+ **wicandidateprofile-ms**

service

write

0/6

Primary candidate ES writer — largest write surface · pinned: models 0.6.0 · wi-elastic toolkit 0.8.0 · click to expand

### wi-candidateevents-ms

- + Candidate activity ES writer — job\_activity\_v2, last\_app\_open, intent\_bucket · pinned: models 1.1.0 · wi-elastic toolkit 0.8.0 · click to expand

service

write

0/6

### wiemployerviewcandidate-ms

- + Candidate ES writer — employer interaction fields · pinned: models 0.6.0 · elastic toolkit 0.5.0 · click to expand

service

write

0/4

### wiemployerjobpost-ms

- + Job ES writer — job feature save/update on go-live · pinned: models 0.7.1 · wielasticsearchadapter 0.2.0 · click to expand

service

write

0/4

### workindiajobs-ms

- + Primary job ES reader — relevancy engine, main list search · pinned: models 0.9.0 · elastic toolkit 0.5.0 · click to expand

service

read

0/6

### wiemployerservicingms

- + Reads candidate + job features for employer servicing flows · pinned: models 0.2.0 · click to expand

service

read

0/3

## New / Infrastructure

### NEW: single-write-consumer

- + P1 draft — unified Kafka consumer (not confirmed); only writer to C1 and C3 if adopted · pinned: new service · click to expand

infra

write

0/5

### NEW: embedding-pipeline

- + P3 critical path — ~9.4M context vectors · pinned: new service · click to expand

infra

write

0/4

### Infrastructure / Ops

- + Cluster provisioning, Consul, Kafka, monitoring · pinned: ES 9.4.2 · click to expand

infra

infra

0/6

## Q&A

15 questions

Search questions and answers...

All

BBQ

C1

C2

C3

C4

ES9

HNSW

MAU

P0

P1

P2

P3

P4

P5

P6

adapter

backfill

bbq\_hnsw

candidate

clusters

consul

consumer

cutover

dense\_vector

draft

drift

dual-write

elasticquery-dsl

elasticsearchtoolkit

fragmentation

freeze

int8\_hnsw

jobs

kNN

kafka

legacy

migration

phased

production

quantization

reads

rollout

scope

semantic

services

timestamp

upgrade

vector

wi-elasticsearchmodels

wielasticsearchtoolkit

Expand all

Collapse all

15 total

## ARCHITECTURE

2

+ What are C1, C2, C3, and C4?

clusters

C1

C2

C3

C4

+ Why do we need to migrate Elasticsearch?

migration

ES9

C1

C2

C3

C4

semantic

vector

kNN

upgrade

## MIGRATION PHASES

3

+ Why migrate 3M MAU first instead of all 47M candidates?

P2

P5

MAU

scope

+ When does production cutover to ES 9 happen?

P6

cutover

consul

+ Do we need consumers ready with dual-write before we freeze writes and start C1→C3 migration?

P2

dual-write

freeze

consumer

C1

C3

## VECTOR SEARCH

1

+ What are HNSW, int8\_hnsw, and BBQ? Why are they required — and what are the trade-offs?

HNSW

int8\_hnsw

BBQ

bbq\_hnsw

dense\_vector

C3

C4

P0

P3

quantization

kNN

## WRITE PATH

4

+ Is P1 (single Kafka consumer) confirmed?

